

# Redox- and bio-activity of apocynin (acetovanillone) in tobacco, a plant phenolic that alleviates symptoms of autoimmune diseases in animals

C. Jacyn Baker<sup>a,\*</sup>, Norton M. Mock<sup>a</sup>, Andrey A. Averyanov<sup>b</sup>

<sup>a</sup> Molecular Plant Pathology Lab. U.S. Department of Agriculture, Beltsville, MD, USA

<sup>b</sup> Research Institute of Phytopathology, B. Vyazemy, Moscow Region, 143050, Russia

## ABSTRACT

Apocynin (APO) was shown to be the major constituent of early herbal medicines used to reduce painful suffering from many ailments. Because of its proven utility in rat models, it is now an active candidate for treating humans for several major autoimmune diseases such as Alzheimer's, arthritis, and cardiovascular diseases. In plant research, APO is referred to as acetovanillone, a secondary metabolite that is induced in the leaf apoplast during bacterial invasion. Since its role in plants is unknown, we examined whether any of the attributes found in medical studies could be found in plants. In medical studies, APO is considered to be an NADPH oxidase (NOX) inhibitor, however when tested in tobacco suspension cells, APO failed to inhibit NOX activity. In addition, when APO was used to pretreat plants against virulent or avirulent pathovars of *Pseudomonas syringae*, there was no reduction of symptoms. When pathovars were incubated with 100  $\mu$ M APO for 90 min, there was only a slight decrease in the bacterial population, suggesting that it was not toxic. In animals, a key step for the inhibition of NOX by APO appears to be its oxidation by myeloperoxidase, yielding dimers and trimers. In vitro studies using pH 6, which is more comparable to the plant environment, APO oxidation by horseradish peroxidase yielded similar products. We monitored the redox potential during some of these experiments, trying to begin looking at this parameter, which like pH, can have a more global effect on closed systems such as the apoplast. Phenolics such as APO that are induced upon bacterial ingress, seem likely to play a role in the plant bacterial interaction, but it will require additional and creative research to find and fully understand this role.

## 1. Introduction

One of the first responses of a plant to bacterial pathogens occurs in the leaf apoplast where different phenolics are induced. Acetovanillone is one of the major phenolics that accumulates. Its concentration varies during different phases of the bacterial infection as well as with the type of interaction, saprophytic, susceptible or resistant. It is redox sensitive and would be affected by the oxidative burst that plants produce in response to their initial detection of bacterial invasion. Chemically, it would affect the redox environment of the apoplast, which is the result of the different concentrations and redox potentials of redox-sensitive molecules within the apoplast. It may also have a bioactive effect, yet to be discovered, on the plant/bacterial interaction. Acetovanillone is referred to as apocynin (APO) in the medical community where reports of its potential benefit for various human-related maladies has increased over the last decade [36]. Because of the numerous medical benefits

that have been attributed to APO, we wanted to examine its effect on plant/bacterial interactions more closely, taking into account the findings of the medical research.

One of the first documented uses of APO was in Canada in 1883 [36]. An extract from the root of Canadian hemp (*Apocynum cannabinum*) was described as helping in remedies for dropsy and heart troubles. One of the main components of the extract was found to be APO. In India during the 1970's, the root of *Picrorhiza kurra*, a native plant, was part of traditional Indian medicine with APO as the major medicinal constituent [38]. Several areas of medicine began to show an interest in APO and demonstrated symptom relief for several diseases, using rats and mice as models. At the core, these studies involved autoimmunity and associated inflammation maladies caused by reactive oxygen species (ROS) overproduction, such as superoxide and H<sub>2</sub>O<sub>2</sub>. The ROS is produced by a membrane bound NADPH-oxidase (NOX). In plants, the NOX homologs have been named respiratory burst oxidase

Abbreviations: APO, apocynin; DIAPO, diapocynin; DPI, diphenyleneiodonium chloride; NOX, NADPH oxidase; PSS, *Pseudomonas syringae* pv. *syringae*; PTAB, P. s. tabaci; ROS, reactive oxygen species; SOD, superoxide dismutase

\* Corresponding author.

E-mail address: [jacyn.baker@ars.usda.gov](mailto:jacyn.baker@ars.usda.gov) (C.J. Baker).

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homologs (Rboh) and they are also involved in ROS production in response to pathogens (Sagi and Fluhr, 2001; Torres et al., 2002). APO is believed to inhibit NOX activity and reduce ROS, and thus reduce damage to the tissue. Despite the numerous publications, > 1500, the exact role and mechanism of APO is still controversial [36,48].

Anti-arthritis activity of APO was one of the first areas that showed positive effects. In the early 1990's, studies with APO were conducted on rats that were immunized with collagen to develop joint inflammation which closely resembled the human condition [43]. The APO-treated rats were given 0.024–16 mg/kg body weight per day in their drinking water for 2 weeks. Treated rats had a significant reduction in joint swelling compared to controls and no flare-up in swelling was observed for 100 days. Interestingly, there was minimal swelling at the lower doses of APO and less protection as the dose increased. The researchers felt that the APO was interacting with the neutrophils, which were attacking the arthritic site in an autoimmune response. The APO blocked the neutrophil's ability to produce ROS, which had been shown to be the key cause of inflammation [21]. Since then many more studies have investigated its use in arthritic relief in rats [25,31,40,47].

APO has also been beneficial in fighting vascular diseases, affecting the circulatory system, lungs and heart [30,33,48]. Healthy vascular endothelial cells produce low levels of ROS from NOX and mitochondrial metabolism; this healthy level of ROS is countered by scavengers. During pathological conditions, however, the ROS production is excessive causing oxidative stress and hypertension. The excess of ROS breaks down nitric oxide, which is a critical vasodilator, into toxic peroxynitrite. Rats that had received apocynin, 30 mg/kg body weight, in their drinking water for 6 weeks, showed improved symptoms. They had reduced arterial pressure compared to controls and several other parameters were also improved, including the nitric oxide levels.

A trimer of APO was found to be beneficial for myocardial infarction, or heart attack, in which the blood flow to the heart is too low causing damage to the heart muscle. This study was carried out in rats by using isoproterenol to induce the infarction with the myocardial damage being similar to that in humans [15,44]. When rats were fed APO, 20 mg/kg per day, whether it was before or after the isoproterenol treatment, they showed improvement in wide range of biochemical, histological, and cardiac markers. The treatment blocked ROS and inflammation, as well as, increased the antioxidant status of tissue. It was found that oxidative products from APO were more active than the monomer in this regard. A dimer, diapocynin, and a trimer of APO was tested and also found to have a hypotensive and vasorelaxant effect on treated rats [35]. The trimer was most effective and was reported to target the cysteine residues on one of the cytoplasmic components of NOX, which prevented the assembly and activation of the oxidase [29,30].

Diapocynin, is being used in studies of neurodegenerative disorders such as Parkinson's, Alzheimer's and Lou Gehrig's diseases [16,23,24,27,37,42,49]. A common feature of these diseases is that certain microglia cells, which normally protect the neurons from invasive materials, become overly activated and produce excessive ROS via an inducible NOX. These chronic conditions cause the neurodegeneration. In a mouse model system of Parkinson's disease, diapocynin, 300 mg/kg, was found to protect neurons in a preclinical mouse study and it was also found to halt disease progression in a chronic mouse study [16].

Although the medical benefits of apocynin in a wide range of inflammatory diseases in rats have been demonstrated, its exact role and mechanism of action is still controversial [13,19,41]. Most agree that APO needs to be 'activated', which involves oxidation by peroxidase and H<sub>2</sub>O<sub>2</sub>. Many suggest that the oxidation products, such as the radical (APO\*), dimer or trimer are the actual active components. More recent evidence suggests that apocynin interacts with the sulfhydryl region of the cytoplasmic components of the enzyme, preventing them from binding to the membrane component. Others suggest that APO itself can scavenge ROS, specifically H<sub>2</sub>O<sub>2</sub>.

While there is great interest in APO in the medical world, it has received very little attention in the plant research, where it is generally referred to as acetovanillone. One study with corn seedlings showed that treatment with 100 µM APO for 8–24 h increased the levels of nitric oxide [45,46]. Increased nitric oxide levels lead to increased UV-B tolerance. They suggest that the APO inhibited arginase activity, increasing the amount of arginine available for NO synthesis.

In our studies of apoplast phenolics in tobacco, we have found that several phenolics accumulate in the leaf apoplast within hours of bacterial inoculation [8]. The same response occurs in cell suspension models [3,9,12]. It is accompanied by an ROS burst, thought to involve membrane bound NOX, as well as several other physiological events including a change in ion fluxes and pH, and leakage of nutrients from the plant cell depending on the bacterial pathogen [10]. APO is one of the prominent induced phenolics in both the apoplast and cell suspension systems. Whole tissue analysis of phenolics has shown that APO is undetectable until the leaf tissue is inoculated with the bacteria [4,6]. Then APO is produced and detected in the inoculated tissue within 2 h [8], and at 5 h 50% of it shows up in the apoplast [5]; therefore, it seems likely to play a role in the plant bacterial interaction. Considering the demonstrated effect this compound can play in medical maladies involving ROS, we wanted to directly characterize and test its properties in a plant system.

## 2. Materials and methods

### 2.1. Chemicals

Apocynin (APO) (3'-methoxy-4'-hydroxyacetophenone), also known as acetovanillone, diapocynin, horseradish peroxidase (P8250), myeloperoxidase (M6908), superoxide dismutase (S5771), diphenyleneiodonium chloride (D2926), were obtained from Sigma-Aldrich Chemicals, Inc (St. Louis, MO 63178 USA). Horseradish peroxidase was assayed with pyrogallol as described by Sigma Chemical Co. One unit of peroxidase will form ~18 µM of purpurogallin from pyrogallol per minute at 25 °C. A superoxide dismutase unit was defined as inhibiting the rate of reduction of cytochrome C by 50% in a coupled system using xanthine and xanthine oxidase at pH 7.8 at 25 °C in a 3.0 mL reaction volume with a xanthine oxidase concentration that will cause a change in absorbance of 0.025 min<sup>-1</sup> at 550 nm. Sigma Flagellin fragment (fig 22) was obtained from Anaspec EGT (Fremont, CA 94555 USA).

### 2.2. Preparation of bacteria

The preparation of bacteria was as previously described [8]. *Pseudomonas syringae* pv *syringae* (PSS), and *P. s. pv. tabaci* (PTAB) were routinely maintained on King's B media, augmented with 25 µg/ml nalidixic acid for PSS. PSS caused tissue collapse and a resistant hypersensitive response (HR) on tobacco after 15 h, while PTAB caused a susceptible response and developed symptoms more slowly by 2–3 days; PF is considered a non-pathogen causing no visible symptoms. For experiments, bacteria were grown for 18–20 h in King's B broth (augmented with nalidixic acid for PSS) in dark, at 190 rpm, 30°C. Overnight cultures were centrifuged at 8000 rpm for 3 min at room temperature in a fixed rotor. Pellets were washed with deionized water, centrifuged, and then suspended in deionized water to various concentrations based on optical density; 0.1 ODU<sub>600nm</sub> was about 10<sup>8</sup> CFU ml<sup>-1</sup>.

Heat-killed bacterial preparations consisted of a 0.1 ODU suspension of *P. s. pv. syringae* which was autoclaved for 20 min.

Dilution-plating was used to estimate bacterial density in samples and was carried out as previously described [6]. Aliquots, 10 µl, of a series of varying dilutions (1:5 or 1:2) in sterile water were plated on King's B agar. Plates were incubated at 30 °C for 24–48 h until colonies were large enough to count.

### 2.3. Preparation of cell suspensions and monitoring redox potential

Tobacco (*Nicotiana tabacum* L. cv. Hicks) suspension cells were maintained and prepared for experiments as previously described [1]. Suspension cells were transferred at 4-day intervals and 2-day old cultures were used for experiments to ensure cells were in log phase. Suspension cells were washed and suspended in assay buffer (0.5 mM MES; 0.5 mM  $\text{CaCl}_2$ ; 0.5 mM  $\text{K}_2\text{SO}_4$ ; 0.175 M mannitol, adjusted to pH 6.0 with 1N NaOH) to a cell density of  $0.05 \text{ g ml}^{-1}$ .

For redox potential assays, 20 ml of plant cells, were added to 50 ml polypropylene beakers, capped with size 10 rubber stoppers containing two 1-cm holes, one for electrode insertion and one for adding treatments, sampling and aeration. Changes in the redox potential of the suspensions were monitored using platinum redox combination (AG/AgCl reference) electrodes and data acquisition adapter from Microelectrodes Inc. (Londonberry, NH, USA). Up to 16 beakers were monitored concurrently on a rotary shaker, 190 rpm and  $27^\circ\text{C}$ , allowing 30 min for equilibration before treatments were added. The potential detected by this technique is a resultant of the redox potential and amount of each interacting redox species present.

### 2.4. Preparation of plant material

Tobacco plants (*Nicotiana tabacum*) cv Hicks were grown as previously described [5] under greenhouse conditions at  $21\text{--}27^\circ\text{C}$ . Plants were routinely used for experiments at approximately 6 weeks of age, having 7–10 fully developed leaves. Leaves were numbered from the top down starting with the first fully developed leaf above the first internode. Leaves 4 and 5 were generally the first fully expanded leaves and used for experiments. The center panels were inoculated with bacteria. The leaf halves served as replicates. Inoculation was carried out by pricking the leaf panel in a few spots with a needle and then gently pressing a plastic needle-less syringe barrel, filled with inoculum, against the leaf, which is supported on the opposite side by hand. Two to three plants were used per experiment and all experiments were repeated at least two times.

### 2.5. Monitoring the redox potential of in vitro reaction mixtures

The redox potential of in vitro reaction mixtures was measured with a redox electrode (Thermo Scientific, No. 9678BNWP combination electrode, Beverly, MA 01915, USA) inserted directly into the mixture. This is a platinum redox sensor with a silver/silver chloride internal reference system. The mV readings from 5 electrodes were recorded every second using a Parstat MC amplifier (Ametek, Kent, OH 44240, USA).

### 2.6. Analysis of phenolics by UPLC/UV/MS

Phenolic analysis was using a ultra-pressure liquid chromatography system (Waters Acquity UPLC, Milford, MA, USA) equipped with a refrigerated autosampler, a photodiode array detector (PDA), and ESI interface and single-quadrupole MS detector (Thermo Fisher Scientific, Waltham, MA, USA). Mass spectrometer settings were ESI probe temperature of  $350^\circ\text{C}$ , scan time  $0.25 \text{ s} \pm \text{mode}$  and cone voltage of 50 V. Samples, 4  $\mu\text{l}$ , were separated on an Acquity BEH C18 ( $2.1 \times 50 \text{ mm}$ ,  $1.7 \mu\text{m}$ ) column using a binary solvent system; A) methanol, B) 0.1% formic acid. The program steps for solvent A were: 0 min, 45%; 5 min, 45%; 10 min, 65%; 12 min, 45%; 17 min, 45%. The solvent rate was 0.15 ml per min and the effluent pH was around 3. The PDA measured absorbance from 210 to 400 nm. Peaks were identified by their retention times and UV spectra. The molecular weight or fragment masses (determined by mass spectrometer) were used to verify comparison of peaks between samples. Standard compounds, apocynin and diapocynin were also analyzed using the same conditions.

Quantification was done by Waters Empower software using peak

height of UV absorbance at the maximum wavelength, between 250 and 400 nm, for each peak. A standard curve for APO, using optical density at 275 nm, was linear through  $100 \mu\text{M}$  at 275 nm. The wavelength max of DIAPO shifted to about 284 nm from which the peak height was determined. The absorbance at  $\lambda_{\text{max}}$  of DIAPO was approximately twice that of APO for equal concentrations. Since we did not have standards of the trimer an assumption was made that it would be approximately 3x that of the monomer, apocynin. The results are represented as ' $\mu\text{M}$  APO equivalents' to more easily follow and account for the APO in each reaction. Therefore, the dimer has 2 equivalents and the trimer has 3 equivalents of apocynin. To obtain their approximate concentrations, the  $\mu\text{M}$  APO equivalents of the dimer and trimer would need to be divided by 2 or 3, respectively.

### 2.7. Reaction mixtures

Reaction mixtures, 10 ml, were contained in 20 ml beakers in a temperature-controlled water bath, stirred by magnetic bars. The reactions generally contained 100–500  $\mu\text{l}$  of substrate and  $\text{H}_2\text{O}_2$  plus 3–20  $\mu\text{l}$  of enzyme in 50 mM phosphate buffer, pH 6.0 or 7.4. Aliquots, 100  $\mu\text{l}$ , were removed periodically for UPLC analysis and redox electrodes continuously monitored the redox potential.

Diapocynin was difficult to dissolve in water and we did not want to use DMSO since we were unsure of its redox properties; therefore, we slowly added 1 N NaOH to a 10 mM stock solution of diapocynin until it dissolved at about pH 10. It was then carefully titrated with 1 N HCl to about pH 7. No buffer was used. When added to reactions mixtures the diapocynin remained dissolved and did not affect the pH.

## 3. Results

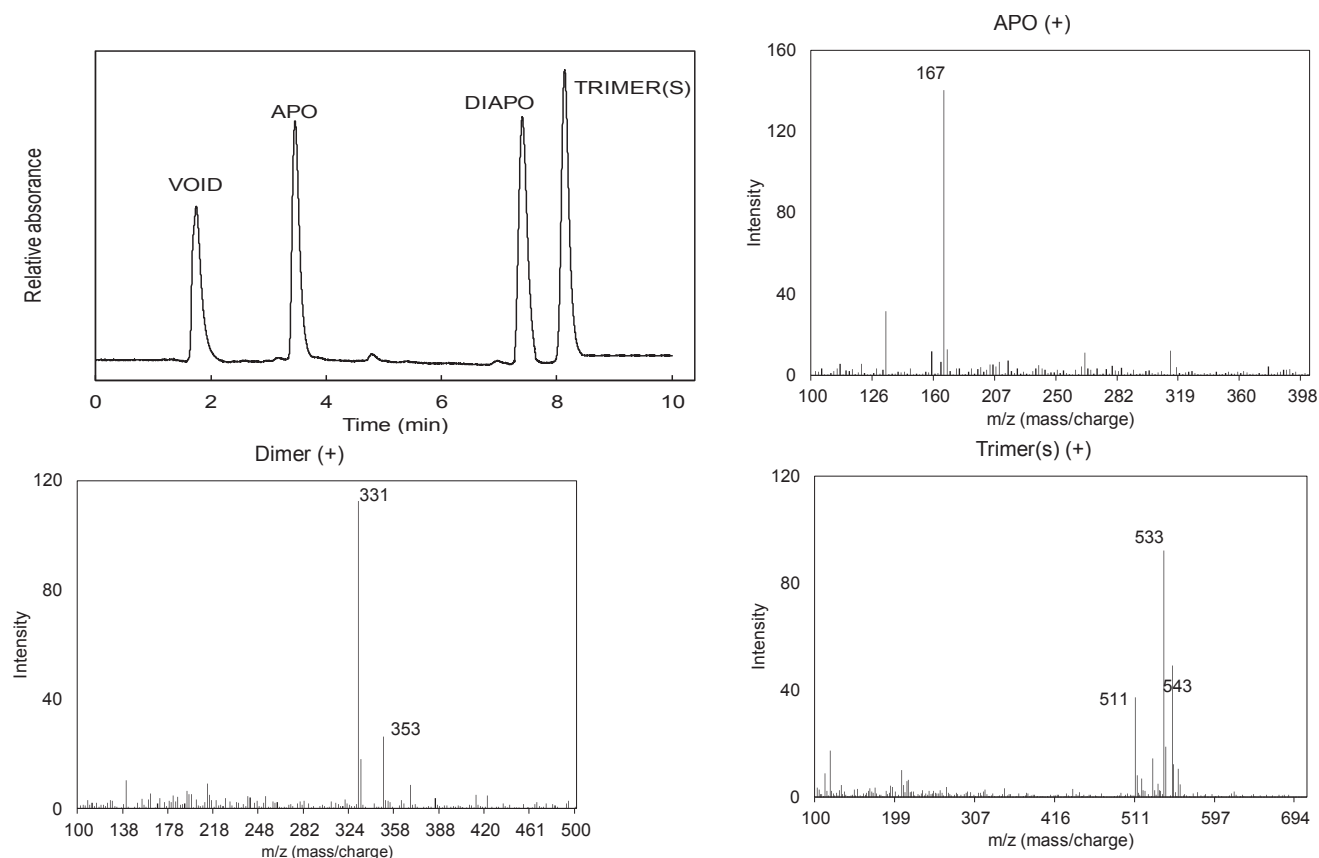
### 3.1. Identification of apocynin oxidation products

The dimer and trimers were the major detectable oxidation products from apocynin. Samples were obtained at different times from reaction mixtures containing 500  $\mu\text{M}$  apocynin, 500  $\mu\text{M}$   $\text{H}_2\text{O}_2$  and 1.8 mU/ml peroxidase at pH 6 over a 90 min period. MS/UV spectra confirmed that the UPLC peaks corresponded to: monomer, 3.6 min; dimer, 7.5 min; trimer, 8.2 min (Fig. 1A). The major mass species from each peak are shown (Fig. 1B,C,D): monomer, RT 3.5 min, mol wt. 166,  $m/z^+$  167; dimer, mol wt. 330,  $m/z^+$  331; trimers, mol wt. 494, 510,  $m/z^+$  511, 533, 543 peaks of mass spectra. The trimers have been shown to have different configurations which are beyond the capability of our equipment and expertise to differentiate, however, they would be expected to have a  $m/z$  species in the range of 500 [14,26,29].

### 3.2. Effect of pH on apocynin metabolism and redox potential

The pH of the plant apoplast, where APO accumulates during bacterial stress, is generally slightly acidic,  $\sim\text{pH}$  6, although it may increase during stress [2,32]. Conversely, the pH of animal cell cytoplasm is around pH 7.2 and blood is around 7.4. We examined the effect of these two environments on APO metabolism. This difference in pH had a significant effect on the UV absorption spectrum of apocynin,  $\sim 100 \mu\text{M}$  (Fig. 2A). The peaks at 230 and 275 nm, decrease as the pH increases from 6 to 8, while an absorbance peak develops near 342 nm. This shift in UV absorbance suggests that the molecular structure has been altered, possibly due to a hydroxyl group, which has a  $\text{pK}_a$  of 8.2 [34]. At pH 7.2, about 10% of the APO would be ionized with an increasing amount as it neared pH 8.2.

To determine the effect pH had on apocynin oxidation, reaction mixtures containing 500  $\mu\text{M}$  apocynin, 500  $\mu\text{M}$   $\text{H}_2\text{O}_2$  and 1.8 mU/ml peroxidase at pH 6 and 7.4 were compared (Fig. 1B and C). The substrate and products were periodically monitored by UPLC-UV. At pH 6, the apocynin was quickly oxidized, nearly 80% within the first 2 min under these conditions, yielding lower levels of the dimer and trimer. At



**Fig. 1.** Separation and mass spectra of apocynin oxidation products on UPLC. A. Compilation of UPLC-UV chromatograms of samples during the oxidation of apocynin. B-D. Mass spectrum using + ev electron spray ionization of: (B) apocynin (mol wt 167), (C) diapocynin (mol wt 330), (D) trimer(s) (mol wt 494, 510).

pH 7.4 the reaction was much slower taking about 30 min for 80% oxidation of the apocynin, with most of the product being the dimer. The pH optimum for the enzyme is between pH 6 and 7, which could account for a small part of the difference in catalytic rate. However, it is more likely the change in the substrate, due to pH, had decreased the success rate of enzyme substrate collisions.

### 3.3. Effect of substrate/ $H_2O_2$ ratio on apocynin oxidation

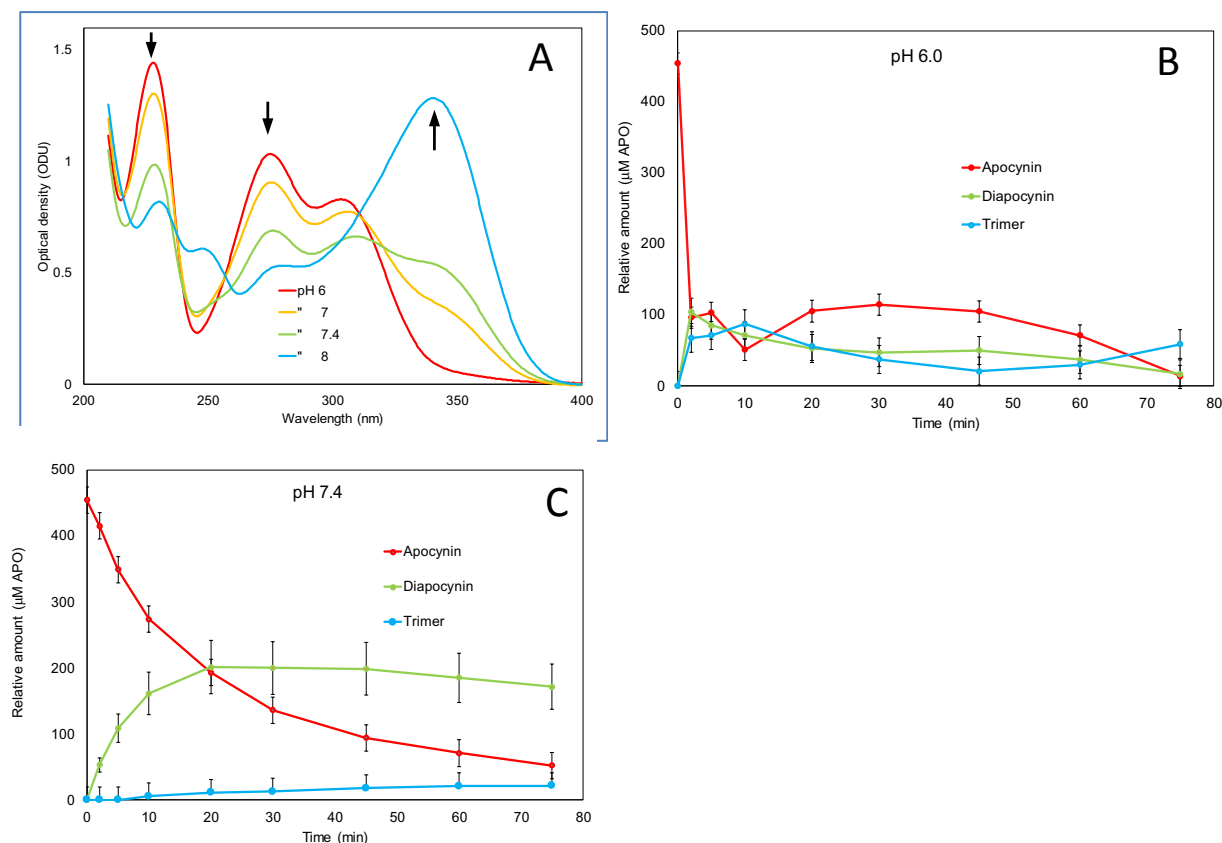
One major line of thought in the medical studies is that apocynin must be activated or oxidized by  $H_2O_2$  and peroxidase to form a product that inhibits the NOX enzyme. Using horseradish peroxidase instead of myeloperoxidase, we examined the oxidation of APO in vitro focusing on the kinetics and the effects on redox potential at pH 6, which is closest to the apoplast. We first noted that the ratio of substrate: $H_2O_2$  affects the ratio of products produced as well as the redox potential (Fig. 3). The reaction mixture contained the same initial amount of APO, 500  $\mu$ M, with differing amounts of  $H_2O_2$ , 100, 500, or 1000  $\mu$ M (ratios of 1:0.2; 1:1, or 1:2 APO/ $H_2O_2$ ). When APO exceeded the  $H_2O_2$ , the reaction stopped when the  $H_2O_2$  was depleted and about half of the APO remained (Fig. 3A); about 150  $\mu$ M APO went to forming the dimer and 30  $\mu$ M went to trimer. The remainder, about 70  $\mu$ M, APO formed several minor products which were not identified or not detected. Theoretically, one molecule of  $H_2O_2$  will oxidize 2 molecules of APO, meaning the 100  $\mu$ M  $H_2O_2$  in the reaction mixture would oxidize 200  $\mu$ M APO, not 250  $\mu$ M. The apparent loss of the extra APO is due to the APO\* radical reacting with unoxidized APO to produce the dimer; 75  $\mu$ M APO\* reacted with an equal amount of APO to form 150  $\mu$ M of dimer. The 30  $\mu$ M of trimer consists of 20  $\mu$ M APO\* and 10  $\mu$ M APO. Therefore, only about 95  $\mu$ M APO\* is accounted for as dimer and trimer.

In the other two reaction mixtures, the APO is almost completely oxidized due to an excess of  $H_2O_2$ , the trimer followed by the dimer are the most prominent products (Fig. 3B and C). The results are similar for both reactions, although, the initial reaction rate is slightly slower with the 1:1 ratio, peaking at 5 min, instead of 2 min. Also the amount of dimer stabilizes earlier, at about 60 min, compared to more than 75 min with the 1:2 ratio. The continued loss of both the trimer and dimer is likely due to the remaining  $H_2O_2$  and/or small amounts of the radical producing additional polymers that were not detected.

The redox potential for these reactions varied (Fig. 3D). The reaction with excess  $H_2O_2$  (Fig. 3C) reached the highest redox level, about 0.44 V and slowly drifted downward. The other two reactions with  $H_2O_2$  equal to or less than the apocynin substrate (Fig. 3B and C), had redox traces were similar to each other. The reaction with the 1:1 ratio increased higher than the 1:0.2 ratio reaction. Another product, which is not detected by this technique, was the fully oxidized form of apocynin. These findings will be discussed in terms of a hypothetical metabolic model in the discussion, section 4.

### 3.4. Effect of enzyme concentration on redox potential

To determine the effect of enzyme concentration on the redox potential during apocynin oxidation, reaction mixtures containing 50  $\mu$ M APO, 25  $\mu$ M  $H_2O_2$ , and 0.6, 1.2, or 1.8 U/ml horseradish peroxidase in 50 mM phosphate buffer, pH 6, were monitored for redox potential (Fig. 4). The main difference was the time to reach the peak potential with the highest enzyme activity having the fastest peak time. This initial increase generally follows the oxidation of the apocynin.



**Fig. 2.** Effect of pH on apocynin UV absorbance, oxidation with peroxidase and  $\text{H}_2\text{O}_2$ , and redox potential. A. UV absorbance of apocynin,  $\sim 100 \mu\text{M}$ , in phosphate buffer at pH 6 or 7.4. B and C) Oxidation of apocynin with  $500 \mu\text{M}$  apocynin,  $250 \mu\text{M}$   $\text{H}_2\text{O}_2$ , and  $1.8 \text{ mU/ml}$  of horseradish peroxidase in phosphate buffer pH 6 (B) and pH 7.4 (C). Periodically, samples were analyzed by UPLC-UV. See section 2 for details.

### 3.5. Co-oxidation of diapocynin

Preliminary testing had shown that diapocynin was either not oxidized or very slowly oxidized by peroxidase and  $\text{H}_2\text{O}_2$ , even though it was produced and then degraded during the oxidation of apocynin (Fig. 3 B and C). To determine if diapocynin could be oxidized (co-oxidized) in the presence of apocynin, we monitored reactions containing each phenolic together and separately. Reaction mixtures contained  $100 \mu\text{M}$  of one or both phenolics, plus  $300 \mu\text{M}$   $\text{H}_2\text{O}_2$ , and  $0.85 \text{ mU/ml}$  peroxidase in  $50 \text{ mM}$  phosphate buffer, pH 6 (Fig. 4). The diapocynin, as measured by UPLC-UV, generally fluctuated around its initial concentration of  $100 \mu\text{M}$ , indicating that it was not oxidized by peroxidase (Fig. 5A). The reason for fluctuation likely relates to its relative poor solubility in water, which could interfere with the actual amount passing through the UPLC. When APO is present with the diapocynin, both were oxidized (Fig. 5B). This is consistent with APO being oxidized and forming the radical  $\text{APO}^*$ , which then can then interact with surrounding phenolics, the most abundant being APO or diapocynin early in the reaction forming diapocynin and trimer, respectively. Two radicals can disproportionate into APO and its fully oxidized product. When APO is oxidized by itself, the radical reacts with APO, which is most abundant, producing the dimer and subsequently the trimer (Fig. 5C).

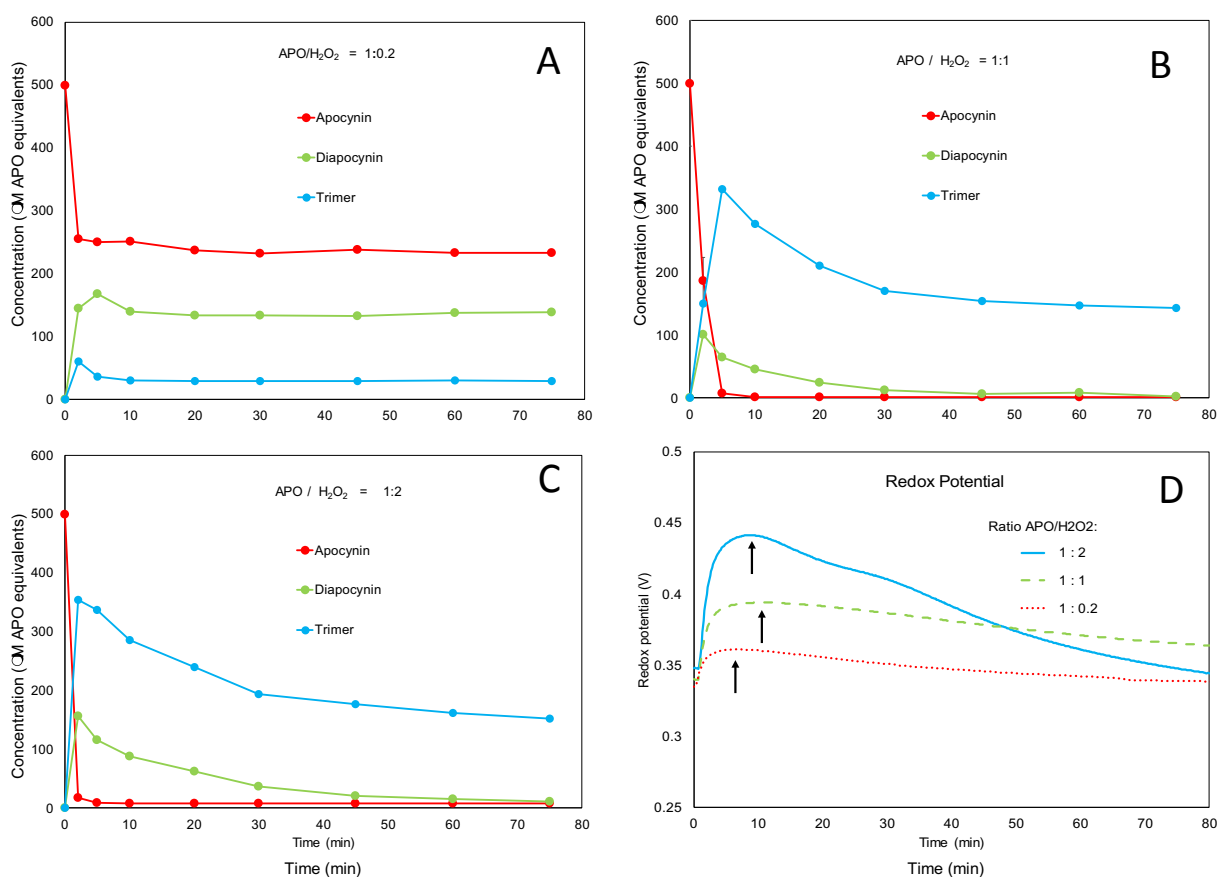
### 3.6. Effect of SOD and hypoxia on kinetics

When phenolic radicals are formed in aerobic conditions, it is feasible that they might react with oxygen donating an electron to form superoxide. To test this, reaction mixtures containing  $250 \mu\text{M}$   $\text{H}_2\text{O}_2$ ,  $0.85 \text{ mU/ml}$  peroxidase and  $500 \mu\text{M}$  apocynin were amended with either SOD ( $\sim 500$  units) or  $\text{N}_2$  gas saturation (Fig. 6). The reactions were

monitored by UPLC-UV and redox, but no obvious difference was detected. If superoxide were being produced during the reaction, it would be expected to dismutate to  $\text{H}_2\text{O}_2$  and  $\text{O}_2$ . The addition of SOD to the reaction would speed up dismutation increasing the  $\text{H}_2\text{O}_2$  concentration and therefore speed up the reaction. Saturation of the reaction mixture with  $\text{N}_2$  would remove  $\text{O}_2$  from the reaction, which would be expected to slow the reaction rate if it had been participating. As seen in Fig. 5 neither of these possibilities occurred.

### 3.7. Effect of APO or DPI on inhibition of the oxidative burst in tobacco cell suspensions

Tobacco cell suspensions were used to test whether APO would inhibit plant NADPH oxidase, as it was reported to do in animals. The oxidative burst caused by the membrane bound NADPH oxidase in these cells was triggered by either flagellin protein or heat-killed bacterial suspension. Tobacco cell suspensions,  $0.05 \text{ g ml}^{-1}$ , were treated with either  $100 \mu\text{M}$  APO or  $3 \mu\text{M}$  DPI, followed by heat-killed bacteria or flagellin protein (Fig. 7A and B). The activation was monitored by the change in the redox potential caused by the oxidative burst. The oxidative bursts caused by either flagellin protein or heat-killed bacteria were similar, increasing about  $0.1\text{--}0.15 \text{ V}$ . In both cases, the addition of DPI, a known inhibitor of NOX, caused an immediate decrease in the redox potential of about  $0.5$  to  $0.1 \text{ V}$ , clearly preventing an oxidative burst. Apo had the opposite effect, increasing the oxidative burst nearly  $0.3 \text{ V}$  in both cases. The APO control along with the other controls (Fig. 7A), had little effect on the redox potential of non-triggered cell suspensions.

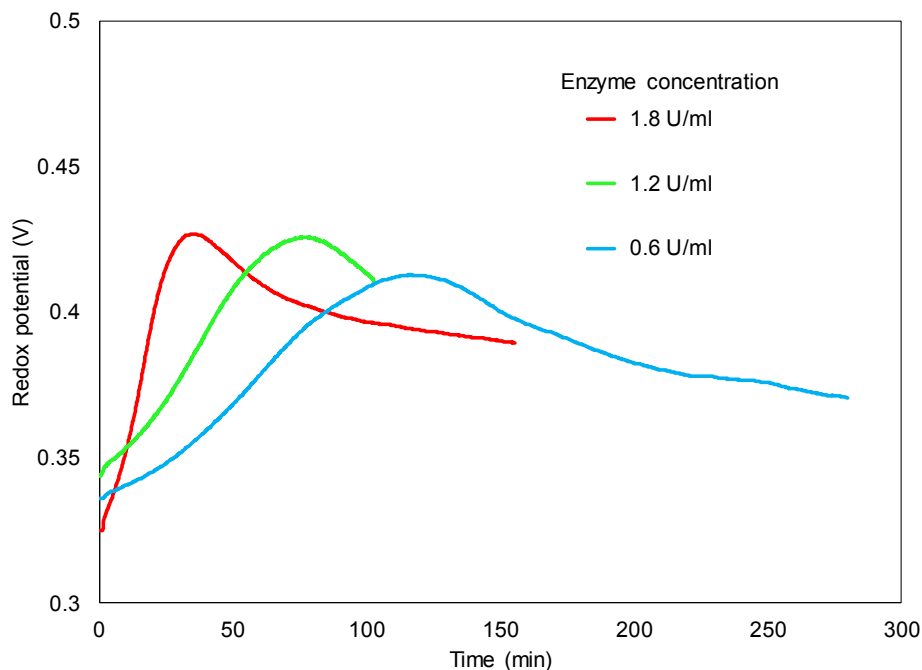


**Fig. 3.** Effect of apocynin:H<sub>2</sub>O<sub>2</sub> ratio on the amounts of the products of oxidation and on the redox potential. The reaction mixture consisted of 500 μM apocynin, 100 μM (A), 500 μM (B), or 1000 μM H<sub>2</sub>O<sub>2</sub> (C), and 3.8 mU/ml of horseradish peroxidase in phosphate buffer, pH 6. Periodically, samples were analyzed by UPLC-UV (A–C), redox potential was monitored continuously (D). See section 2 for details.

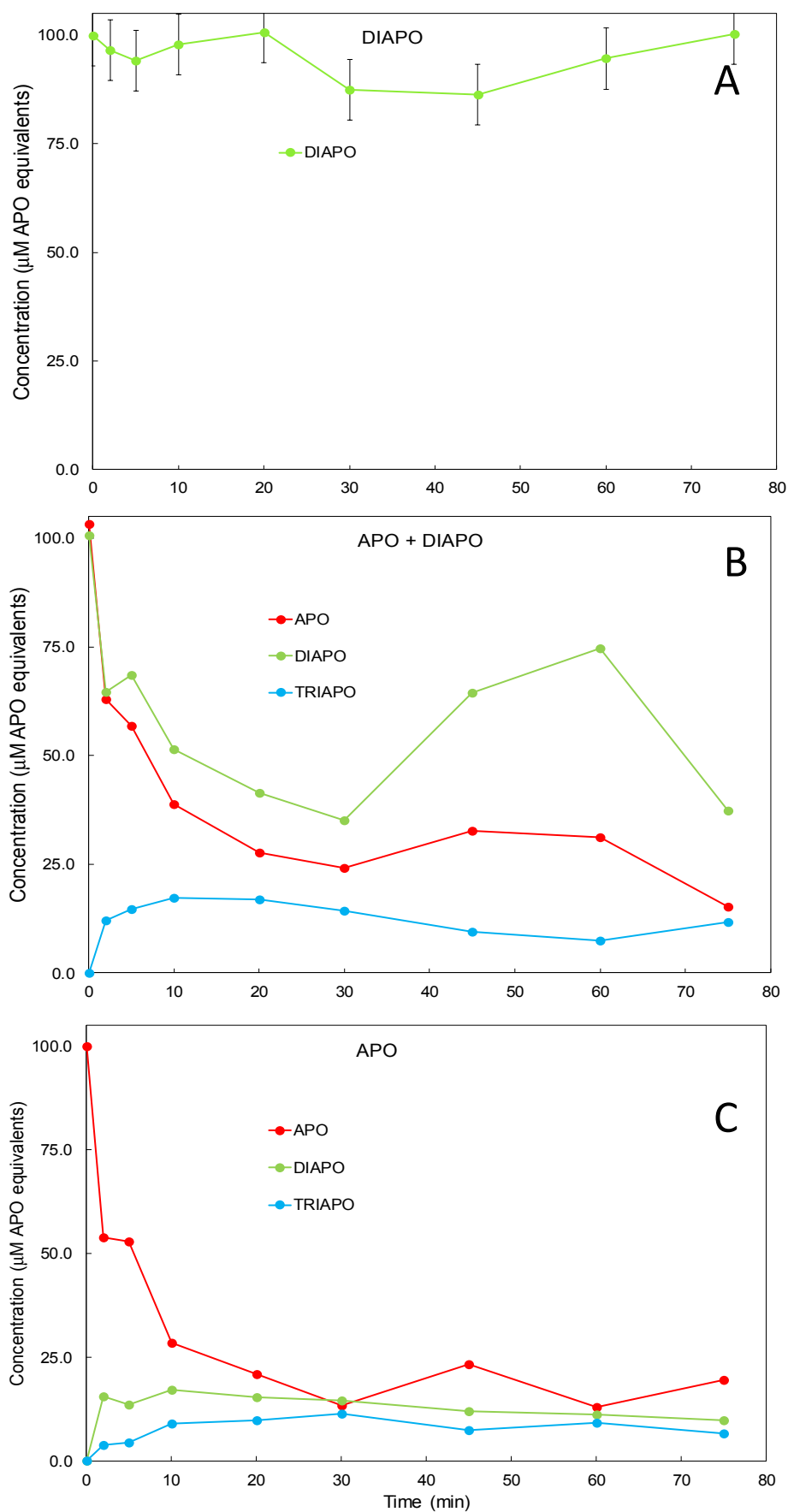
### 3.8. Effect of APO on bacteria and disease symptom development in tobacco

To test the effect of APO or its reaction products on bacteria

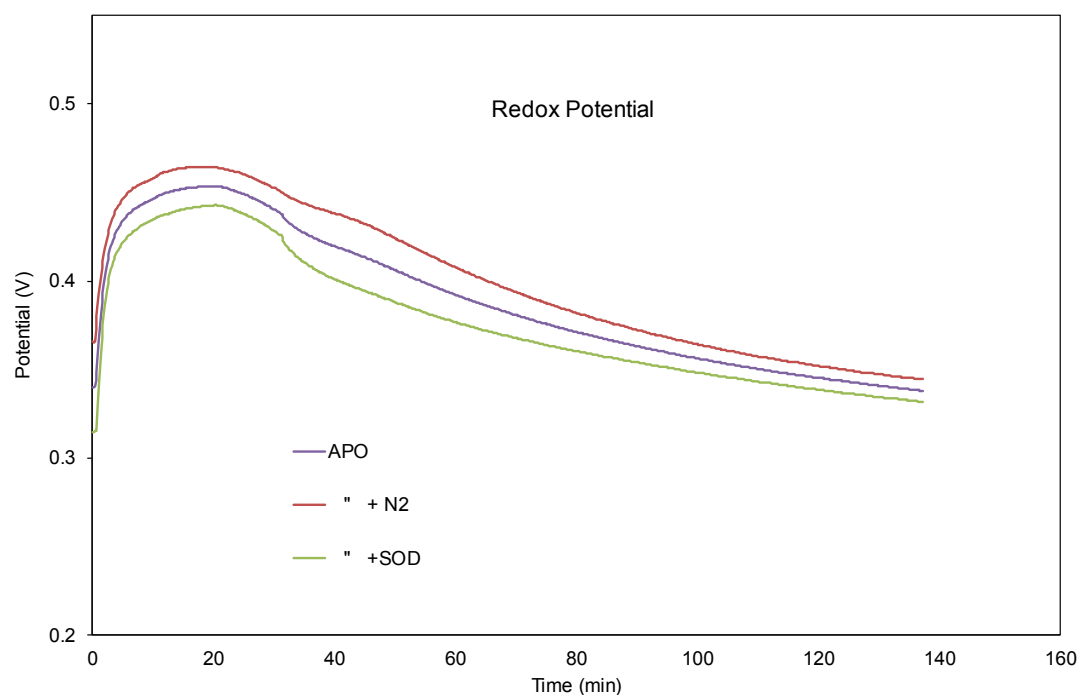
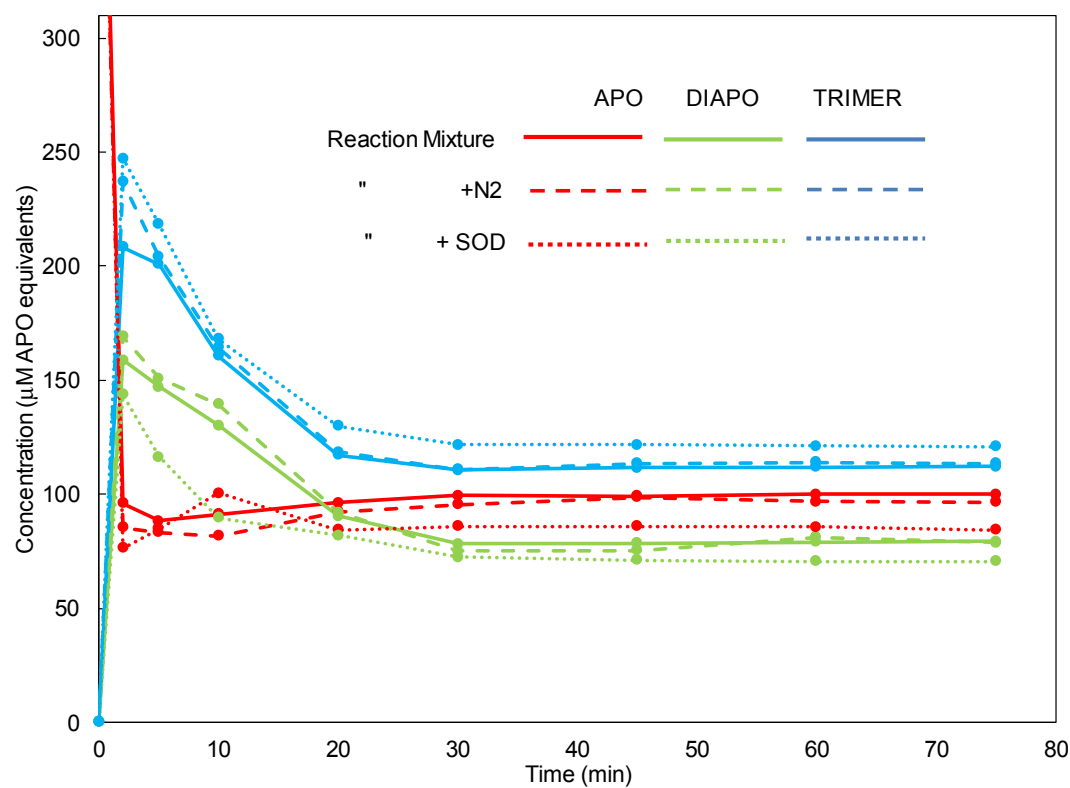
directly, *Pseudomonas syringae* pv. *syringae*, 10<sup>6</sup> cfu ml<sup>-1</sup>, was incubated in a mixture containing 100 μM APO, 100 μM H<sub>2</sub>O<sub>2</sub>, and 4 mU/ml of horseradish peroxidase phosphate in buffer, pH6. Control bacteria



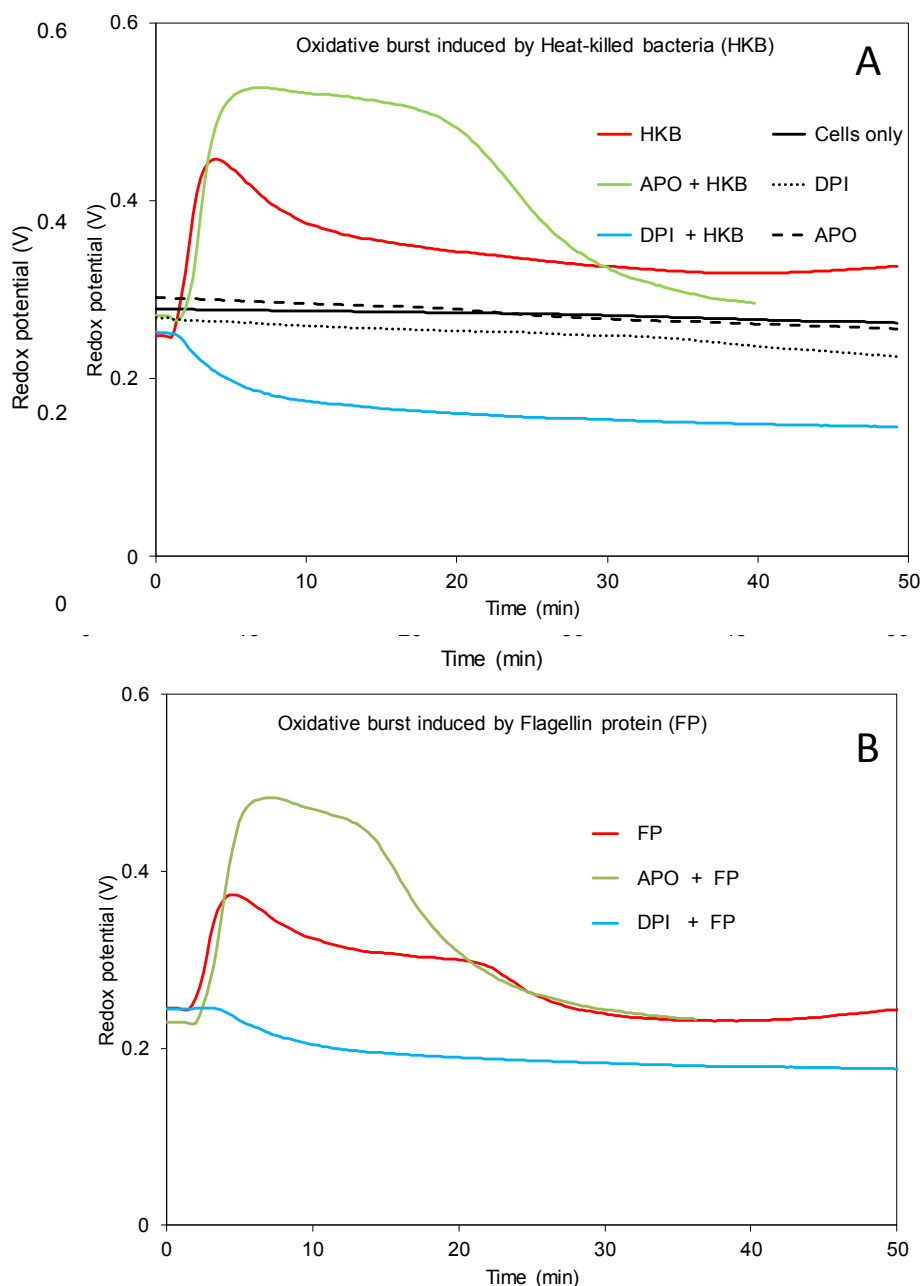
**Fig. 4.** Effect of enzyme concentration on the redox potential during oxidation of apocynin. The reaction mixtures contained 50 μM APO, 25 μM H<sub>2</sub>O<sub>2</sub>, and 0.6, 1.2, or 1.8 mU/ml horseradish peroxidase in 50 mM phosphate buffer, pH 6. Redox potential was monitored continuously. See section 2 for details.



**Fig. 5.** Co-oxidation of diapocynin with apocynin. Reaction mixtures in 50 mM phosphate buffer, pH 6, contained 300 μM H<sub>2</sub>O<sub>2</sub>, 0.6 mU/ml peroxidase plus (A) 100 μM diapocynin, (B) 100 μM each of apocynin and diapocynin, and (C) 100 μM apocynin. Periodically, samples were analyzed by UPLC-UV. See section 2 for details.



**Fig. 6.** Comparing the effects of SOD or N<sub>2</sub> on the oxidation of APO. Reaction mixtures in 50 mM phosphate buffer, pH 6, contained 250  $\mu\text{M}$  H<sub>2</sub>O<sub>2</sub>, 1.2 mU/ml peroxidase and 500  $\mu\text{M}$  apocynin. Some reactions had nitrogen gas (+N<sub>2</sub>) bubbled through the mixture continuously, other mixtures had ~500 units SOD added (+SOD). Periodically, samples were analyzed by UPLC-UV. Redox potential was monitored continuously. See section 2 for details.



**Fig. 7.** Comparison of APO and DPI as inhibitors of the oxidative burst caused by NADPH oxidase in tobacco cell suspensions. Tobacco cell suspensions,  $0.05 \text{ g ml}^{-1}$ , were pretreated for 10 min with either  $100 \mu\text{M}$  APO or  $3 \mu\text{M}$  DPI. Suspensions were then treated with either (A) heat-killed bacteria (HKB) or (B) Flagellin protein (FP) to initiate the oxidative burst. See section 2 for details.

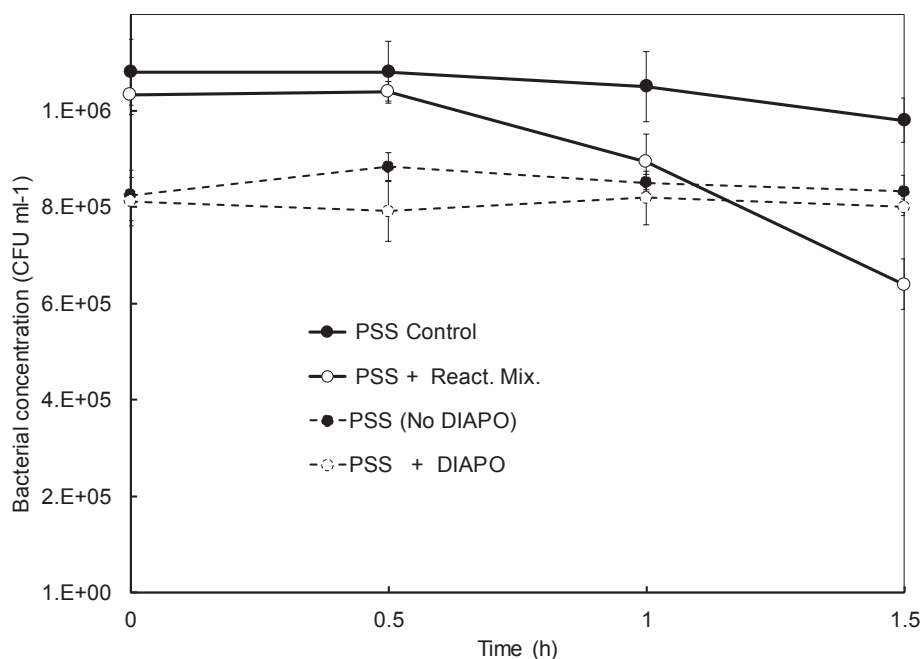
were incubated in only buffer. Over a 90 min period, the bacterial concentration of bacterial suspension was determined every 30 min by dilution plating (Fig. 8). The control bacterial concentration dropped about 10% for the 90 min period. The bacteria incubated in the reaction mixture decreased an additional 15% in concentration for a total of about 25% from  $1 \times 10^6$  to  $7 \times 10^5 \text{ cfu ml}^{-1}$ . The results indicate that the reaction mixture was not highly lethal to the bacteria.

APO was tested for its possible interference with disease development in plants, which like animal systems is thought to involve ROS from membrane bound NADPH oxidase. Tobacco leaf panels were infiltrated with either water or  $100 \mu\text{M}$  APO. After 4 h, the same panels were inoculated with either *P. s. pv. syringae* and *pv. tabaci*,  $10^8 \text{ cfu ml}^{-1}$ . No difference between symptoms of controls and treated plants was found over a 7 day period. The incompatible hypersensitive response with *P. s. pv. syringae* began y 24 h, while the compatible

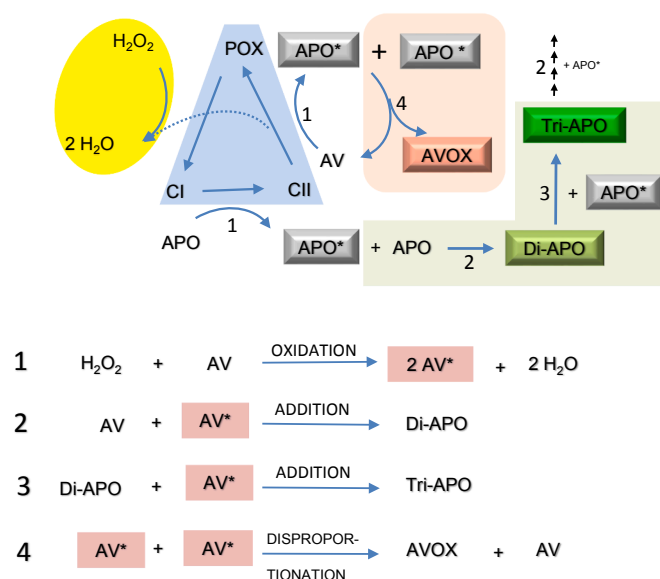
response with *P. s. pv. tabaci*, began by 48 h. There was no visible effect of APO on water inoculated control plants.

#### 4. Discussion

We wanted to better characterize APO in the plant realm and determine if it had properties similar to those found in animal research. An important difference between the animal and plant environment is the pH, which is about 7.4 for blood and 6.0 for the apoplast. There was a clear difference in the UV absorbance of APO (Fig. 2A) at the 2 pHs indicating a likely change in its topographical features. This could account for the slower reaction rate at the higher pH. In this study, the rationale for the differences in reaction products due to pH or changes in proportions of reaction mixture ingredients can be more easily explained using the hypothetical model in Fig. 9. At pH 6 (Fig. 2B), rapid



**Fig. 8.** Effect of APO oxidation and DIAPO on *Pseudomonas syringae* pv. *syringae* viability. For APO oxidation, *P. s. pv. syringae*,  $10^6$  cfu ml $^{-1}$ , was incubated for 90 min in a reaction mixture containing 100  $\mu$ M APO, 100  $\mu$ M H $_2$ O $_2$ , and 4 mU/ml of horseradish peroxidase in phosphate buffer, pH6. Controls consisted of bacteria incubated in buffer. For DIAPO exposure, bacteria were incubated with 100  $\mu$ M DIAPO for 90 min. Bacterial viability was measured every 30 min and the bacterial concentration was determined by dilution plating. See section 2 for details.



**Fig. 9.** Hypothetical model for APO oxidation with H $_2$ O $_2$  and peroxidase. The H $_2$ O $_2$  ultimately is reduced to water, taking 2 electrons from the peroxidase (yellow). The 2 oxidized forms of peroxidase, CI and CII (blue), each gain an electron from APO leaving it with a single electron in the outer most molecular orbit, APO\*. These APO\* radicals have a higher redox potential, but are somewhat stable due to their phenoxyl structure and can interact with molecules. Interaction with APO can lead to polymerization (green) leading to a dimer, Di-APO, followed by a trimer, Tri-APO. Interaction with another APO\* can lead to disproportionation (red) leading to APO and APOOX, the fully oxidized product.

oxidation of APO would yield a relatively high concentration of APO\* that would favor polymerization into dimer which is then converted to trimer. At pH 7.4 (Fig. 2C), due to the slower reaction rate, the concentration of APO\* remains lower and interacts with the APO which remains at a relatively high concentration. The dimer is the major product until its concentration increases and it competes for APO\* and is converted to the trimer.

There was controversy in the animal studies over whether the APO itself was the bioactive molecule or whether the dimer or trimer

oxidation products were responsible for the reduction of the autoimmune symptoms. During apocynin oxidation, we found that due to the involvement of the APO\* radical and its ability to interact with several reaction components, the final outcome varied depending on the pH, enzyme activity, and the ratio of APO and H $_2$ O $_2$ . In the latter case, the different results due to the different reactant ratios (Fig. 3) could also be explained by the model discussed above (Fig. 9). If the reaction has more APO than H $_2$ O $_2$  (Fig. 3A), it is likely that each APO\* radical will react with APO, the most abundant molecule surrounding it, followed by the dimer. Once the H $_2$ O $_2$  is gone the activity is over and the concentrations of reactants remains the same. If the ratio of the two reactants is similar (Fig. 3B and C) such that the H $_2$ O $_2$  is capable of oxidizing much of the APO and soon be in competition with APO\* for unoxidized APO. This brings the APO to near zero meaning very little APO\* is produced. In these reactions, the dimer and trimer continued to degrade until the dimer is depleted. This could be due to H $_2$ O $_2$  still being present and in unknown reactions causes further breakdown. Regarding the redox potential (Fig. 3D), the higher amounts of H $_2$ O $_2$  likely resulted in higher levels of APO\* reflected by the higher potentials. After this peak the potentials return to the equilibrium potential, about 0.35 V.

Several studies had reported that apocynin needed to be activated by oxidation with H $_2$ O $_2$  and peroxidase [38,40,41], leading some researchers to test the reaction products directly, avoiding the need for ‘activation’. Diapocynin was tested in several studies and found to be similar or better than apocynin leading some to suggest that diapocynin replace apocynin in these studies [20,22,26,35,39,50]. However, an organic chemistry study examined the reaction mixture from apocynin oxidation further finding dimers through heptamers, and concluded the trimer was most inhibitory to NADPH oxidase. None of these studies considered the kinetics of the reaction. We examined diapocynin and found it to be unreactive with peroxidase and H $_2$ O $_2$  (Fig. 5A), meaning that if it were substituted for apocynin no other products, such as the trimer, would be formed. Interestingly, we found that the dimer was co-oxidized if apocynin was present (Fig. 5B). This suggests that the APO\* radical that is formed is possibly the major radical leading to the polymerization of apocynin as shown in Fig. 9.

Our studies of apocynin on plant and bacterial did not find the same effects found in the animal studies. When testing for inhibition of plant NADPH oxidase, we used tobacco suspension cells. Plants also have

membrane bound NADPH oxidases which will produce an oxidative burst during bacterial invasion. Previous work with these suspension cells had demonstrated that they could recognize and respond appropriately to resistant, susceptible and saprophytic bacteria [1,3,7,10,11,17,18]. We have found the oxidative burst can be monitored by the redox potential, which coincides with a period of increased oxygen uptake, hydrogen peroxide production, and during which time a pool of accumulated phenolics were oxidized. Phenolics, similar to those induced in leaves, are induced upon recognition of the bacteria. They respond to bacteria or bacterial components with an oxidative burst much like the animal response. However, while DPI inhibited the oxidative burst as measured by redox potential, the apocynin did not (Fig. 7). Apocynin caused an increase in the redox potential, suggesting that its product was a prooxidant and would likely magnify the oxidative burst.

In a previous study we had shown that acetosyringone, another induced phenolic found in the apoplast, would cause PSS to stop multiplying almost instantly without killing them [28]. We tried the same treatment of PSS with 100  $\mu$ M APO or DIAPO with little effect; about 10–15% loss in viability of PSS after 90 min of exposure to APO and no loss with DIAPO (Fig. 8). The inhibitory effect was accumulative over the time period rather than nearly instant as found with our previous work with acetosyringone.

A major contention in the animal work was that APO itself did not appear to directly inhibit NOX, that it required ‘activation’ by oxidation with a peroxidase and hydrogen peroxide [38,42,43]. From this work, they found a dimer of APO, and eventually a trimer [19,22,29,30]. We found similar products using a plant peroxidase (Figs. 1–4). However, as can be seen from some of graphs, products obtained from apocynin oxidation depends on many circumstances, such as, pH, level of enzyme, the initial amount of APO, and more.

As mentioned above APO is a prominent induced phenolic in both the tobacco leaf apoplast and cell suspension model systems. Its role remains unclear. When we found out that acetovanillone, as we referred to it, had been and still is being investigated in the medical field under the name apocynin (~2000 articles) we wanted examine it under further. We did learn much more than we knew about its oxidation, which likely occurs in the apoplast, but in the presence of other phenolics. Unfortunately, the bioactivity aspects, such as blocking NADPH oxidase, interfering with plant pathogen, or bacterial toxicity were not detected in our study. We monitored the redox potential with some of these experiments trying to begin looking at this parameter, which like pH can have a more global effect on closed systems such as the apoplast. Induced phenolics such as APO seem likely to play a role in the plant bacterial interaction, but will require additional and creative research to find and fully understand this role.

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